

ASSIGNMENT ON CSE - 260

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Answer to the Question No. 1

(a)

$(101110010001)_2$ needs to be converted to decimal

$$\begin{aligned} \Rightarrow & 1 \times 2^{11} + 0 \times 2^{10} + 1 \times 2^9 + 1 \times 2^8 + 1 \times 2^7 + 0 \times 2^6 \\ & + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ = & 2961 \end{aligned}$$

(b)

$(111011.101)_2$ needs to be converted to decimal

$$1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \rightarrow \text{Whole number part}$$
$$= 27$$

fraction part $\begin{array}{ccccccc} 1 & 1 & -1 & -2 & -3 & 1 & 1 \\ & & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}$

$$= 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} \rightarrow 0.625$$

$$= 27.625$$

Answer to the Question No 2

$(4195)_{10}$

2	4195	→
2	2097	→ 1
2	1048	→ 1
2	524	→ 0
2	262	→ 0
2	131	→ 0
2	65	→ 1
2	32	→ 1
2	16	→ 0
2	8	→ 0
2	4	→ 0
2	2	→ 0
2	1	→ 0
	0	→ 1

→ 1 0 0 0 0 0 1 1 0 0 0 1 1

Answer to the Question No. 3

$$(513)_{10}$$

$$\begin{array}{r|l} 16 & 513 \\ \hline 16 & 32 \rightarrow 1 \\ 16 & 2 \rightarrow 0 \\ & 0 \rightarrow 2 \end{array} \quad \uparrow$$

So, hex will be 201

Answer to the Question No. 4

(a)

$$(29)_{12} \rightarrow (?)_7$$

First we convert $(29)_{12}$ to decimal.

$$2 \times 12^1 + 9 \times 12^0 = 33 \quad \therefore \text{So, it is } (33)_{10}$$

Now, we convert $(33)_{10}$ to $(?)_7$

$$\begin{array}{r|l} 7 & 33 \\ \hline 7 & 4 \rightarrow 5 \\ & 0 \rightarrow 4 \end{array} \quad \rightarrow (45)_7$$

(b)

$$(10110111)_5 \text{ to } (?)_4$$

First we convert $(10110111)_5$ to decimal.
and then decimal to $(?)_4$

$$(1 \times 5^7 + 0 \times 5^6 + 1 \times 5^5 + 1 \times 5^4 + 0 \times 5^3 + 1 \times 5^2 + 1 \times 5^1 + 1 \times 5^0)$$

$$\rightarrow (81906)_{10}$$

$$\text{so, } (81906)_{10} \text{ to } (?)_4$$

$$\begin{array}{rcl} 4 \overline{) 81906} & & \\ 4 \overline{) 20476} & \rightarrow & 2 \\ 4 \overline{) 5119} & \rightarrow & 0 \\ 4 \overline{) 1279} & \rightarrow & 3 \\ 4 \overline{) 319} & \rightarrow & 3 \\ 4 \overline{) 79} & \rightarrow & 3 \\ 4 \overline{) 19} & \rightarrow & 3 \\ 4 \overline{) 4} & \rightarrow & 3 \\ 4 \overline{) 1} & \rightarrow & 0 \\ 0 & \rightarrow & 1 \end{array}$$

$$\text{so, Ans} \rightarrow (10333302)_4$$

Answer to the Question No. 5

$(412)_9$ and $(134)_9$

Addition

$$(412)_9 \longrightarrow (335)_{10}$$

$$\begin{array}{r} (134)_9 \longrightarrow (112)_{10} \\ \hline (546)_9 \longrightarrow (447)_{10} \end{array}$$

$$\begin{array}{r} 9 \overline{) 447} \\ 9 \overline{) 40} \longrightarrow 6 \\ 9 \overline{) 5} \longrightarrow 4 \\ 0 \longrightarrow 5 \end{array}$$

So, $(546)_9$ is verified

Subtraction

$$\begin{array}{r} (412)_9 \longrightarrow (335)_{10} \\ - (134)_9 \longrightarrow (112)_{10} \\ \hline (267)_9 \longrightarrow (223)_{10} \end{array}$$

$$\begin{array}{r} 9 \overline{) 223} \\ 9 \overline{) 24} \longrightarrow 7 \\ 9 \overline{) 2} \longrightarrow 6 \\ 0 \longrightarrow 2 \end{array} \longrightarrow (267)_9$$

Multiplication

$$(412)_9$$

$$(335)_{10}$$

$$(334)_9$$

$$(1112)_{10}$$

$$\hline 1798$$

$$1336 \times$$

$$412 \times \times$$

$$\hline 57448$$

$$56418$$

Now, let's verify,

$$\begin{array}{r} 9 \overline{) 37520} \longrightarrow 8 \\ 9 \overline{) 4168} \longrightarrow 1 \\ 9 \overline{) 463} \longrightarrow 4 \\ 9 \overline{) 51} \longrightarrow 6 \\ 9 \overline{) 5} \longrightarrow 5 \\ 0 \end{array}$$

So, it is $(56418)_9$, [verified]

Answer to the Question No. 6

$$(+27) + (-13)$$

$$27 \rightarrow 0011011$$

$$13 \rightarrow 1101$$

$$+13 \rightarrow 01101$$

$$\rightarrow \text{bits} \rightarrow 0001101$$

$$1110010$$

$$+1$$

$$2's \rightarrow 1110011$$

Complement

$$\text{Now, } 27 \rightarrow 0011011$$

$$-13 \rightarrow 1110011$$

$$\underline{11001110}$$

We don't include carry bit, so $\rightarrow 0001110$

$$\text{So, } 14 \rightarrow (27-13) \rightarrow 0001110$$

We are adding numbers of two different sign

and got a '+' . So, No overflow,

$$OF = 0$$

Answer to the Question No 7

Add 13 and 27 in 6 bits $\rightarrow (4+27) + (+13)$

27 $\rightarrow$ ¹0 ¹1 ¹0 1 1

13 $\rightarrow$ 0 0 1 1 0 1

1 0 1 0 0 0

$\swarrow$ signifies negative

We know summation of 2 positive values we would have '0' in front, but there is 1. So, there is overflow.

OF = 1

Answer to the Question No. 8

(a)

Here,

$$(91)_{10} = (0101011)_2$$

$$499 = (111110011)_2$$

$$15 \text{ bit} \rightarrow 000\ 0000\ 01011011$$

$$000000\ 11111011$$

$$1's \text{ comp} \rightarrow 111\ 1111\ 10100100$$

$$\rightarrow 111\ 111\ 00000100$$

+1

+1

$$2's \text{ comp} \rightarrow 111\ 1111\ 10100101$$

$$111\ 111\ 00000101$$

Now, by one's complement,

$$91 - 499 = (+91) + (-499) \rightarrow$$

$$000000\ 001011011$$

$$111111\ 000001100$$

$$(-) \leftarrow \underline{111111\ 001100111}$$

So, accordingly we get no. overflow:

Now, by 2's complement

$$91 = 499 \rightarrow$$

$$000\ 0000\ 01011011$$

$$111\ 1110\ 00001000$$

$$\underline{111111\ 0011}$$

Now, by 2's complement

$$91 - 499 = 000\ 0000\ 010\ 11011$$

$$111\ 1110\ 000\ 01101$$

$$111\ 1110\ 01101000$$

Negative
sign

Here, when a large value is subtracted from a smaller value we are supposed to get a negative sign

$$OF = 0$$

(b)

$$(379)_{10} = (101111011)_2 \quad (98)_{10} = (110\ 0010)_2$$

$$15\text{-bit} \rightarrow 000\ 000\ 101111011$$

$$00000000\ 110\ 0010$$

$$000\ 000\ 101111011$$

$$00000000\ 110\ 0010$$

$$111\ 111010000100$$

$$11111111\ 0011101$$

+1

+1

$$1111\ 11010000101$$

$$11111111\ 00111110$$

$$30, \text{ Adding } (+379) + (+98) \rightarrow$$

